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Artificial intelligence in food system: Innovative approach to minimizing food spoilage and food waste

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ABSTRACT

Globally, about one-third of all food produced for human consumption is lost or wasted, compounding issues of food security, economic inefficiency, and environmental harm. Artificial Intelligence (AI) presents transformative potential to mitigate these losses by enhancing food spoilage predictions and optimizing supply chain management. This paper examines the deployment of AI technologies such as machine learning models, predictive analytics, and advanced algorithm in predicting food spoilage with high accuracy, thereby reducing food waste substantially. Key innovations highlighted include early detection systems for spoilage indicators, dynamic algorithms for optimal storage conditions, and predictive models for waste forecasting based on real-time environmental data. A review of case studies, including AI-driven solutions from Shelf Engine and Afresh, shows a 14.8 % reduction in food waste per store, with an associated reduction of 26,705 tons of CO_2 emissions. Similarly, IKEA achieved a 30 % reduction in kitchen food waste within one year using AI-powered monitoring systems. Despite these successes, challenges in data collection, model training, and the integration of AI into existing food management systems persist. These include issues related to data quality, legacy system compatibility, and regulatory barriers. The paper concludes with actionable recommendations for future research, urging interdisciplinary collaboration to develop standardized data protocols, enhance real-time monitoring capabilities, and address the ethical implications of AI adoption in the food sector. By advancing these strategies, AI's full potential in curbing global food waste can be realized.

1. Introduction

The World Food Programme (WFP) reports that one-third of food produced worldwide for human consumption is wasted at various stages of the food supply chain [1]. This waste is enough to feed two billion people, a critical fact given that 30 % of the world's population currently experiences moderate to severe food insecurity, and over 900 million people face severe food shortages [2]. Food waste not only exacerbates these issues but also affects food availability by influencing market demand and supply dynamics, which in turn can drive up prices during periods of scarcity. Furthermore, food waste contributes to broader societal issues by intensifying economic disparities. Those who are wealthier can better afford food, thus widening the gap between social classes. In addition, the environmental impact of food waste is significant, as decomposing waste emits harmful pollutants. To address these multifaceted problems, solutions to reduce food waste are essential. Artificial Intelligence (AI) has emerged as a potent tool in this regard, particularly through its ability to predict food spoilage before it

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happens, potentially reducing waste significantly. By leveraging AI, there is potential to improve the sustainability of the global food supply chain, presenting a pathway to a more stable and equitable future.

Every year, approximately 1.3 billion tons of food are wasted worldwide, resulting in financial losses estimated at US$1 trillion [3]. Table 1 offers a detailed summary of food waste distribution and its primary causes throughout various stages of the supply chain. Chauhan et al. [4] identify that approximately 24 % of food waste occurs at the production and post-harvest stages, attributed mainly to inefficient agricultural practices and inadequate storage facilities, respectively. In addition, 17 % of waste arises during transportation, often due to improper handling and adverse environmental conditions. At the retail level, 20 % of food is wasted due to overstocking and stringent cosmetic standards. However, the consumption stage accounts for the most significant loss, 35 %, predominantly influenced by consumer behaviors such as excessive purchasing.

The economic impact of food waste is profound, encompassing not only the direct loss of food value but also indirect costs including healthcare expenses, pollution, and waste management [9]. These economic repercussions vary based on household and national economic status, with wealthier entities generally wasting more food at the consumer level, whereas poorer households and developing countries experience higher losses post-harvest [10].

Food waste not only highlights the disparities between affluent and impoverished populations by affecting their respective food access and security [11] but also imposes substantial environmental costs. The cultivation of food demands extensive use of critical resources such as land and water, which are squandered when the food is not consumed [6]. In addition, the disposal and decomposition of wasted food contribute significantly to environmental degradation, emitting an equivalent of 3.3 Gt of carbon dioxide annually [6]. These environmental and economic impacts underline the urgent need for strategic interventions to mitigate food waste at all levels of the supply chain.

This paper seeks to review the broader impacts of AI applications on reducing economic losses due to food waste and minimizing environmental damage through more effective resource management and waste reduction strategies. We aim to address the following research questions.

RQ 1: How can artificial intelligence (AI) technologies be applied to predict and reduce food spoilage across different stages of the food supply chain?
RQ 2: What are the challenges associated with integrating AI into existing food management systems, and how can these challenges be overcome to improve the efficiency and effectiveness of food spoilage prevention?
RQ 3: What potential does AI hold for improving economic and environmental sustainability in the food supply chain through enhanced spoilage prediction and waste reduction?

Table 1
Overview of food waste by stage in the supply chain.

Stage	Percentage of Total Waste	Key Factors Contributing to Waste
Production	24 %	Inefficient farming techniques, natural disasters
Post-Harvest	24 %	Poor storage facilities, lack of proper infrastructure
Transportation and Retail	17 %	Inadequate transportation methods, delays, exposure to unsuitable environments Overstocking, cosmetic standards, improper handling
Consumption	35 %	Over-purchasing, lack of planning, poor consumer habits

Source: Adapted from Chauhan et al. [4], WFP [3], Xue et al. [5].

These questions investigate the deployment of AI tools such as machine learning models and predictive analytics in enhancing the accuracy of food spoilage predictions, which could lead to significant reductions in food waste. We explore the obstacles related to data collection, model training, and the practical integration of AI technologies into the food industry's operational frameworks, emphasizing the need for standardized data and interoperable systems.

This study provides a holistic analysis of AI applications across multiple stages of the food supply chain, integrating real-time monitoring, predictive modeling, and decision-support systems. Unlike previous studies, this research explores how AI can dynamically adapt to real-world complexities, including fluctuating food demand, variable storage conditions, and multi-stakeholder interactions. The study presents a novel framework for AI-driven food spoilage detection and waste reduction, leveraging machine learning, IoT, and blockchain technologies to enhance decision-making and efficiency.

2. Methodology

2.1. Research approach

A comprehensive literature review was conducted to establish a foundational understanding of AI applications in minimizing food waste. To ensure a broad yet rigorous examination of existing research, multiple databases were utilized, including PubMed (for studies related to food systems), Google Scholar (to access a wide range of scholarly articles, including grey literature), and Scopus (for interdisciplinary research coverage).

2.2. Search strategy

The search strategy incorporated Boolean operators (AND, OR, NOT) to refine results and improve relevance. Keywords included "AI in food systems", "food spoilage prediction", "machine learning in food waste management", "supply chain optimization", and "AI-driven food sustainability".

2.3. Eligibility criteria

Articles were selected based on the following criteria.

- Inclusion Criteria: Relevance to AI-driven food waste reduction, publication in peer-reviewed journals, and the presence of empirical data or robust theoretical frameworks.
- Exclusion Criteria: Studies lacking methodological rigor or those that did not directly address the application of AI in food waste reduction.

The review primarily covered research published between 2020 and 2024 to capture recent advancements, with an emphasis on the latest studies up to May 2024. This approach ensured that the paper's analysis was grounded in the most current and relevant findings

3. AI innovations in predicting food spoilage

The integration of AI, along with other Industrial 4.0 technologies like big data and Internet of Things (IoT), is anticipated to significantly enhance the food supply chain. According to Romanello and Veglio [12], these technologies are set to improve food quality and safety, boost environmental sustainability, and increase operational efficiency across all stages of the food supply chain. AI is utilized in various phases such as crop monitoring and early pest detection during the planting and production stages, storage monitoring post-harvest, routing optimization in transportation, inventory management in retail, and expiry tracking at the consumption stage. This comprehensive application of AI not only addresses quality and efficiency but also plays a crucial role in reducing food spoilage and waste, offering a potential solution to a persistent

global issue. Fig. 1 outlines the various stages of the food supply chain and the specific applications of AI and other technologies at each stage to enhance efficiency and reduce waste.

In the food supply chain, AI technologies play a critical role across various stages to ensure efficiency and food safety. At the production stage, AI is employed for crop monitoring, pathogen detection in agriculture, and detecting microorganisms in poultry farms, aiming to enhance the quality and yield of agricultural products by preventing diseases and pests [13,14]. For instance, deep learning models have been successfully implemented in crop disease detection, significantly improving early identification and prevention strategies [15]. Moreover, AI-driven spectral imaging techniques have demonstrated high accuracy in detecting pathogens in poultry farms, improving biosecurity measures [16]. During the post-harvest stage, AI aids in monitoring storage environments and sanitizing processing equipment to reduce losses from spoilage and contamination [17]. Studies show that AI-integrated IoT systems enable real-time monitoring of storage conditions, helping to predict and prevent spoilage [18,19].

In transportation, AI applications like routine optimization, temperature control, and cold-chain break analysis ensure perishable goods are transported efficiently and remain unspoiled, particularly vital for temperature-sensitive products [20]. Research indicates that AI-powered predictive analytics enhance cold-chain logistics, minimizing temperature fluctuations that compromise food quality [21]. At the retail level, AI supports inventory management and food safety training to manage stock effectively and ensure compliance with food safety standards. AI-based demand forecasting techniques have improved inventory optimization, reducing food waste in bakeries [22]. Additionally, AI-driven training systems enhance food safety compliance by providing interactive learning modules for food handlers [23,24]. Finally, during consumption, AI helps identify unsafe foods through analysis of online review comments and tracking expiration dates, thus assisting consumers in avoiding spoiled or unsafe products. Sentiment analysis of online reviews has proven effective in detecting foodborne illness outbreaks, allowing for timely intervention [25]. Moreover, AI-powered smart labeling systems aid consumers in tracking product freshness and expiration dates, reducing household food waste [26]. Overall, Fig. 1 highlights the interconnected role of advanced technologies in optimizing each stage of the food supply chain, aiming to minimize waste and enhance food safety from production to consumption. These technological interventions not only aim to increase the efficiency of these processes but also contribute to sustainability by reducing the environmental impact associated with food waste and spoilage.

4. AI for early detection

Food spoilage results from various factors including the presence of pathogens, chemical and biochemical reactions, physical damage, and enzymatic activities. These factors alter the pH, nutritional content, taste, texture, color, and water activity of food, making it unsafe or unpleasant for consumption [27]. AI and machine learning technologies offer promising solutions to mitigate food waste by predicting spoilage with high accuracy before it occurs. These machine-learning models are trained on extensive datasets that include information about food characteristics, ideal storage conditions, and signs of spoilage [28]. These advanced models not only predict the shelf life of food items but also use environmental data such as humidity, temperature, and ventilation to recommend optimal storage conditions, thus preventing spoilage. In microbiology, AI is also repurposed for the early detection of pathogens in food, contributing to food safety by predicting and reducing the risks of foodborne illnesses [29].

Further, AI and big data (characterized by high volume, velocity, and variety of information) play crucial roles in identifying early warning signs of food safety risks during production, such as harmful algal blooms affecting seafood or fungal growth in crops, which could lead to mycotoxin formation. Big data technologies, including machine learning algorithms, cloud computing, and predictive analytics, process vast amounts of real-time data from diverse sources such as IoT sensors, satellite imaging, and microbiological analysis to detect contamination patterns, optimize food distribution, and improve risk assessment in food safety. These technologies enable the food industry to effectively monitor and assess quality and safety in real-time, ensuring product integrity and consumer health [30]. These AI-enabled systems have been effectively implemented in various practical scenarios, such as the US FDA's use of AI to detect problematic seafood imports and vegetable growers using AI to manage crop risks based on environmental factors [31,32]. AI does not only enhance food safety but also reduces food waste by allowing for more precise management of food quality and safety risks throughout the supply chain.

5. AI for optimizing food storage and distribution

AI technology enhances food preservation by predicting and preventing spoilage, aligning closely with sustainable agricultural practices. Notably, a machine learning model described by Sonwani et al. [33] assesses whether fruits and vegetables are stored under optimal conditions by analyzing images of produce under varying temperature and humidity settings using convolutional neural networks (CNNs). This model not only predicts the shelf life of these items but also supports the development of intelligent inventory systems that minimize waste through real-time adjustments of storage conditions. The model uses image recognition algorithms to detect subtle visual cues of spoilage such as color change and texture degradation and pairs this data with environmental sensors to refine predictions of produce deterioration. Additionally, AI's predictive analytics are utilized to anticipate consumer demand patterns, allowing for better alignment of food production with consumption, thus reducing the likelihood of overproduction [34]. Recurrent neural networks (RNNs), such as Long Short-Term Memory (LSTM) models, are employed to recognize seasonal

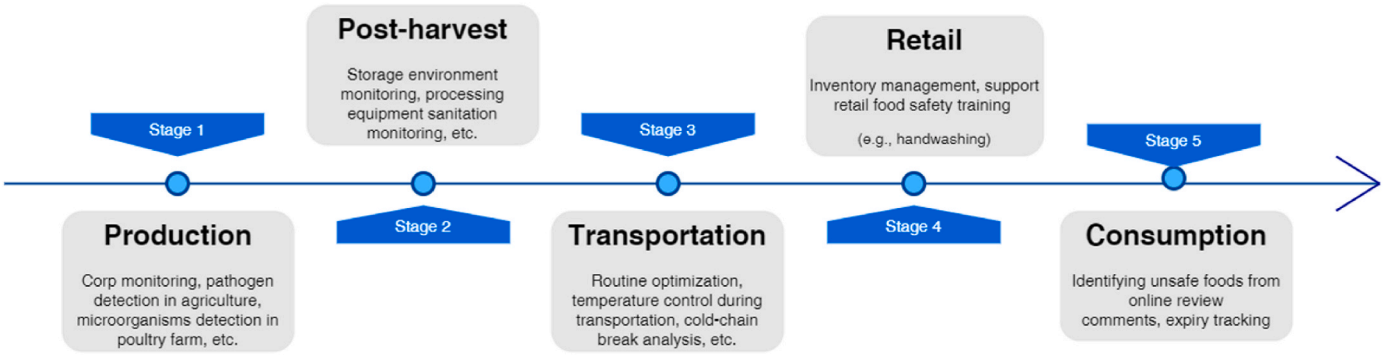


Fig 1. AI application in food supply chain.

fluctuations and forecast future demand, allowing for better alignment of food production with consumption, thus reducing the likelihood of overproduction [35]. These capabilities extend throughout the supply chain, ensuring proper storage conditions are maintained by adjusting factors such as temperature and humidity in real-time. AI also facilitates the efficient redistribution of food to areas with high demand, such as shelters, by integrating logistics optimization algorithms that prioritize deliveries based on freshness and need.

AI also plays a crucial role in managing microbiological risks during storage, transportation, and consumption, by analyzing environmental conditions and identifying patterns that could lead to foodborne illnesses. [24]. Predictive models based on support vector machines (SVMs) or random forests can flag potential risk factors by correlating storage conditions with microbiological data to anticipate outbreaks of spoilage or contamination [24]. These models not only improve food safety but also reduce waste due to spoilage and contamination.

Beyond food safety, AI applications offer significant environmental and economic benefits. For instance, Bhatia et al. [6] report that AI implementation in food management has notably reduced CO_2 emissions, with one case showing an annual reduction of 26,705 tons. Economically, AI-driven forecasting tools, such as demand prediction models and dynamic pricing algorithms, have proven effective in the grocery sector, reducing costs related to unsold inventory and disposal while boosting profitability. The Pacific Coast Food Waste Commitment [8] cites an average profit increase of 14.8 % per store following the adoption of AI technologies for inventory management (see Table 2). These advancements highlight the multifaceted benefits of AI, underscoring its impact on both environmental sustainability and economic efficiency in the food industry.

6. Challenges in implementing AI for food spoilage prediction

Implementing AI for predicting food spoilage presents substantial challenges, primarily stemming from the need for diverse, high-quality data essential for training effective predictive models. Data collection involves accumulating detailed information on variables such as temperature, humidity, storage conditions, and microbial activity. However, acquiring extensive and representative datasets is often difficult due to the variability across different food types, storage conditions, and spoilage processes, as highlighted by Anwar et al. [36]. Each type of food has unique properties and degradation paths, requiring specifically tailored datasets for accurate modeling [37].

Moreover, ensuring the quality and consistency of the data collected is a significant challenge. Data sources vary widely, ranging from sensor outputs and manual observations to historical records, each offering differing levels of accuracy and reliability [38]. Issues such as errors, biases, or missing entries in these datasets can severely impact the performance of AI models, leading to flawed predictions and unreliable outcomes [39].

Another obstacle is the standardization of data collected from disparate sources. The food industry lacks universally accepted protocols for data collection, which complicates efforts to standardize data, a crucial step for developing robust AI applications [38]. These challenges underscore the complexities involved in deploying AI for food spoilage prediction and highlight the need for concerted efforts to improve data quality and standardization in this field. Table 3 provides a summary of key challenges and potential solutions in implementing AI for food spoilage prediction.

Table 2
Financial and environmental impact of AI in food waste reduction.

Impact Type	Quantitative Benefit	Reference Case
Reduction in CO_2 Emissions	Reduction of 26,705 tons annually	Case Study involving AI implementation in retail grocery [8]
Cost Savings	Decrease in costs associated with food disposal and unsold inventory	Grocery retail sector through AI forecasting tools [7]
Increased Profitability	The average increase in profits by 14.8 % per store	AI-enhanced inventory management in multiple stores [8]

Sources: Bhatia et al. [6] for environmental impact figures, and Ciccullo et al. [7] and Pacific Coast Food Waste Commitment [8] for financial benefits.

The integration of AI in food spoilage prediction faces significant hurdles, not least of which is data accessibility and sharing. Ethical concerns about data privacy, proprietary interests, and adherence to regulatory standards frequently obstruct the free exchange of valuable data resources [53]. The reliance on expansive and diverse sets of training data, sourced from various devices such as sensors, cell phones, and cameras, introduces additional complexities related to data security and the potential exposure of sensitive personal information [40]. Moreover, collaborative projects often encounter obstacles in formulating data-sharing agreements, adhering to legal frameworks, and reconciling stakeholder concerns over data ownership and intellectual property rights [41].

To navigate these challenges, implementing strategies for data standardization, fostering open data platforms, and cultivating partnerships is essential. This includes harmonizing data formats, ontologies, and metadata to ensure consistency and interoperability across diverse sources [54]. Open data initiatives, which promote transparency and protect privacy through anonymization and encryption, are crucial in this context [39]. Moreover, collaboration among industry stakeholders, academic researchers, and regulatory authorities is vital to facilitate data sharing, establish best practices, and address the legal and ethical dimensions of data utilization [55].

A notable example of an innovative solution to these issues is the decentralized privacy-preserving technology used by Gorji et al. [40]. This approach leverages federated learning alongside new fluorescence imaging technology and two deep learning models to detect invisible residues on food preparation equipment and surfaces, classifying them as clean or contaminated. This model operates without the need to share data across clients or store data on a centralized server, thus enhancing client data privacy and addressing one of the critical challenges in deploying AI in food safety applications.

As AI and machine learning models become more prevalent in the field of food spoilage, issues of bias in these technologies have also come to the forefront. Ruben, Emilio, and Javier [42] highlight that biases can manifest at various stages of the AI development pipeline, from data collection to model training and deployment. During the data collection and labeling phase, biases may arise if the dataset is incomplete, non-representative, or if different annotators interpret labels inconsistently, leading to skewed outputs from AI algorithms. Similarly, biases can occur in the model training phase if the data is unbalanced or if the model architecture is ill-equipped to handle diverse inputs.

Moreover, it is crucial to continuously monitor and test AI models with a variety of inputs post-deployment to detect and mitigate biases. Thomas et al. [48] note that such biases can exacerbate existing social inequalities if biased data reinforce pre-existing prejudices. This issue is illustrated by Supriya [43], who observed that AI models can display discriminatory patterns against people of color, a problem that often goes unrecognized when the training data itself harbors biases such as out-group homogeneity. These biases not only distort the outputs but also perpetuate and intensify social inequalities and discrimination. Therefore, addressing these biases is essential for ensuring fairness and equity in AI applications within food safety and beyond.

The conceptual framework illustrating sources of bias in AI deployment within food systems (Fig. 2) critically underscores the multifaceted nature of biases that may compromise the accuracy, fairness, and applicability of AI-driven food spoilage prediction models. Data-related biases, such as incomplete or skewed datasets, can arise from over-reliance on historical sales records that fail to capture emerging consumption trends, leading to inaccurate demand forecasting. Algorithmic

Table 3
Summary of key challenges and solutions in implementing AI for food spoilage prediction.

Category	Challenge	Description	References
Data Collection	Diverse, High-Quality Data	Difficult to collect comprehensive data on temperature, humidity, microbial activity across food types.	Anwar et al. [36], Deng et al. [37]
	Data Variability	Different food types require tailored datasets, making data collection complex.	Anwar et al. [36]
	Data Quality Issues	Inaccurate or missing data from sources like sensors or records affect AI predictions.	Wang et al. [38], Voulodimos et al. [39]
	Data Standardization	Lack of universal protocols for data collection complicates consistency.	Wang et al. [38]
Data Accessibility & Sharing	Privacy & Security Concerns	Ethical and legal barriers hinder data sharing, especially sensitive personal data.	Qian et al. [24], Gorji et al. [40]
	Legal & Regulatory Barriers	Intellectual property and legal concerns block data sharing agreements.	Houser and Bagby [41]
	Decentralized Data Solutions	Federated learning can address privacy issues by keeping data decentralized.	Gorji et al. [40]
Bias in AI Models	Data-Related Biases	Skewed or incomplete datasets result in biased predictions.	Ruben et al. [42]
	Algorithmic Biases	Overemphasis on certain factors (e.g., temperature) reduces prediction accuracy.	Ruben et al. [42]
	Operational & Human Biases	Subjective decisions based on intuition affect AI model deployment.	Supriya [43]
Integration & Operations	Legacy System Compatibility	Difficulty integrating AI with existing food management systems.	Patel et al. [44]
	Organizational & Workforce Barriers	Training staff and adapting workflows for AI adoption can be challenging.	Richey et al. [45]
	Financial Constraints	High initial costs and maintenance expenses create financial barriers.	Rejeb et al. [46]
Regulatory & Legal Barriers	Lack of Regulatory Frameworks	AI technologies face regulatory gaps in food safety standards and assessment.	Thakkar et al. [47]
	Data Harmonization	Need for standardized data formats for international collaboration.	Mu et al. [31], Thakkar et al. [47]
Ethical & Fairness Issues	Bias Amplification	AI models may perpetuate societal biases if trained on biased data.	Thomas et al. [48], Supriya [43]
	Ensuring Fairness	Continuous monitoring of AI models is needed to detect and correct emerging biases.	Thomas et al. [48]
Strategies for Solutions	Standardizing Data Protocols	Develop universal protocols for data collection (e.g., FoodON) to improve data consistency.	Dooley et al. [49]
	Active Learning & Synthetic Data	Use active learning and synthetic data to fill gaps and reduce bias.	Shorten and Khoshgoftaar [50]
	Federated Learning	Implement federated learning to protect data privacy while training models.	Sheller et al. [51]
	Continuous Model Monitoring	Use fairness-aware algorithms to monitor and correct biases in deployed AI models.	Mehrabi et al. [52]

biases emerge when AI models disproportionately weight specific spoilage factors such as temperature fluctuations while neglecting variables like microbial growth, thereby reducing predictive robustness. Operational and human biases further exacerbate inefficiencies, as decision-makers may either overtrust or disregard AI recommendations based on subjective intuition, potentially undermining the system's efficacy. Moreover, external constraints, including regulatory inconsistencies and economic disparities, hinder equitable AI adoption, particularly among small-scale retailers who may lack the financial resources to implement sophisticated predictive analytics. For instance, while large grocery chains can leverage AI-driven demand forecasting tools to optimize inventory management, smaller vendors relying on traditional methods could face increased food waste due to less precise stock adjustments. Addressing these biases necessitates a comprehensive approach that integrates diverse datasets, refines algorithmic transparency, and ensures equitable access to AI technologies across the food supply chain, ultimately fostering a more sustainable and efficient food management ecosystem.

To address challenges in AI-driven food spoilage prediction, targeted strategies must be implemented to enhance data standardization, mitigate bias, and improve integration. First, establishing universal data collection protocols is essential for ensuring consistency and comparability across different datasets. Standardized ontologies, such as the FoodON framework [49], can facilitate interoperability and improve data quality. Additionally, techniques like active learning and synthetic data generation can help compensate for gaps in real-world datasets, reducing bias and improving model generalization [50]. To address ethical concerns surrounding data sharing, federated learning where AI models are trained across decentralized devices without sharing raw data has been successfully used in medical imaging and can be adapted to food safety applications [51]. Moreover, ongoing monitoring of AI models using adversarial testing and fairness-aware algorithms can detect and correct biases that emerge post-deployment [52]. Industry collaborations, including partnerships between regulatory agencies, food producers, and technology developers, can also foster the adoption of best practices and regulatory frameworks that balance innovation with ethical considerations [56]. Implementing these solutions will not only enhance the accuracy and fairness of AI models but also support a more sustainable and equitable food supply chain.

Achieving high accuracy, integrity, and reliability in AI models is critical for their effective application in predicting food spoilage. This requires comprehensive training datasets that encompass a variety of food types, storage conditions, and spoilage factors. Jarray et al. [57] emphasize the importance of improving data collection methodologies, advancing feature selection techniques, and refining algorithmic approaches to boost prediction accuracy. Additionally, real-world validation of AI models is vital to ensure their robustness and applicability across different operational environments. This validation process involves rigorous testing in actual settings to evaluate model performance, pinpoint limitations, and adjust algorithms as necessary ([57]; Wang

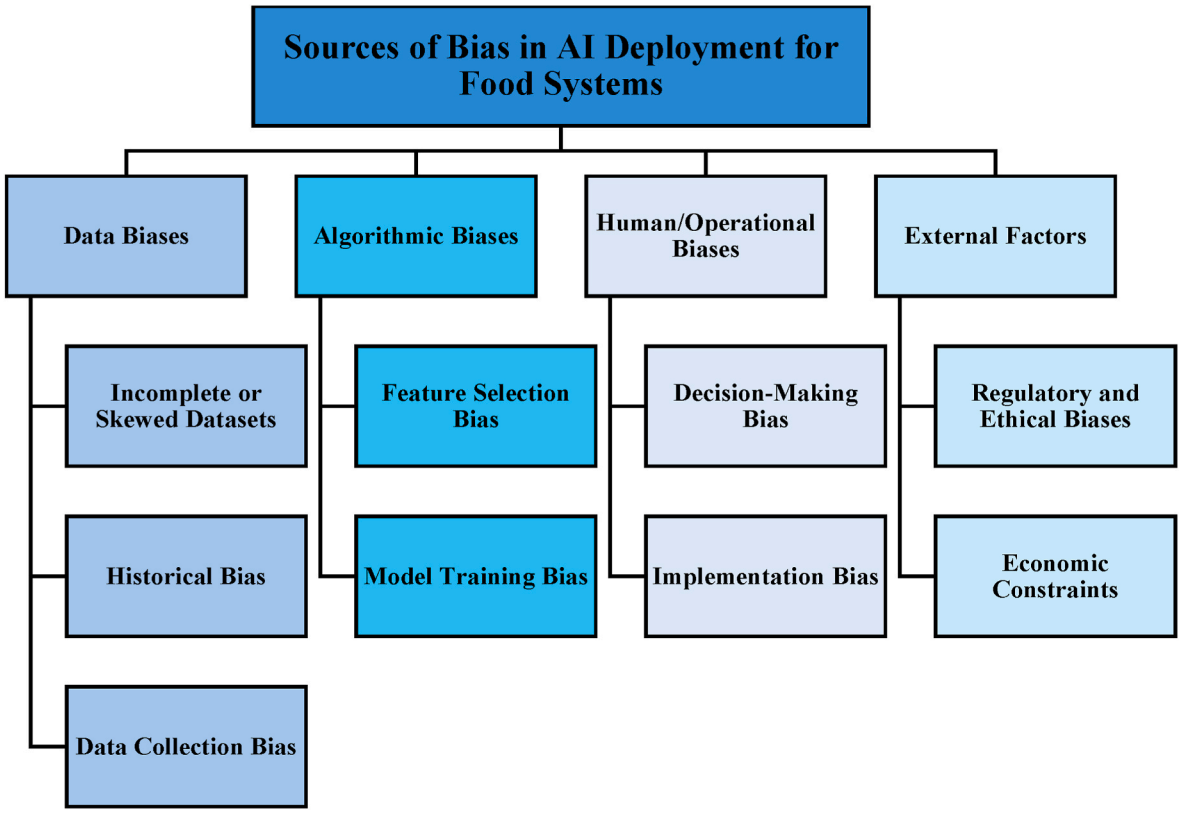


Fig 2. Sources of bias in AI deployment for food systems.

et al., 2021).

Integrating AI technologies into existing food management and supply chain systems also presents significant challenges. Technological issues such as compatibility with legacy systems, interoperability concerns, and the necessity for specialized infrastructure to support AI implementations are notable barriers [44]. Furthermore, operational hurdles include the need to adapt existing workflows and processes to integrate AI insights seamlessly, requiring organizational commitment, workforce training, and change management strategies [45]. Financial obstacles also arise, including substantial initial investment costs, ongoing maintenance expenses, and uncertain returns on investment [46].

Additionally, the regulatory landscape is yet to fully adapt to the integration of AI, with a current lack of appropriate scientific knowledge and assessment practices within regulatory agencies to effectively address the challenges posed by AI technologies [47]. To facilitate international collaboration and data exchange, researchers suggest harmonizing data formats and establishing cooperative platforms and databases, which could help mitigate some of these challenges [31,47]. These efforts are crucial for harnessing AI's potential in improving food safety and reducing waste effectively.

7. Case studies: success stories of AI in reducing food waste

The integration of AI technologies in the grocery retail sector represents a pivotal strategy for reducing food waste, enhancing operational efficiency, and mitigating environmental impacts [7]. These AI systems utilize sophisticated algorithms to process extensive datasets—including sales records, historical trends, and external influences like weather patterns and seasonal variations—to accurately predict consumer demand [58]. This capability allows retailers to optimize inventory levels and minimize over-ordering, thus reducing the incidence of unsold or spoiled products and, consequently, shrinkage and food waste. According to the Pacific Coast Food Waste Commitment [8], the financial advantages of avoiding disposal of unsold inventory and reducing shrinkage highlight the effectiveness of AI in boosting profitability while addressing food waste.

Further emphasizing the operational benefits, AI automates labor-intensive tasks such as ordering and restocking, leading to increased efficiency and cost reductions. Successful pilot programs and the subsequent broader implementation of AI solutions across multiple stores demonstrate the scalability and effectiveness of these technologies in the grocery retail industry.

Several case studies illustrate the concrete impact of AI in reducing food waste. For example, retailers using AI solutions from Shelf Engine and Afresh saw an average food waste reduction of 14.8 % per store, achieving not only reduced shrinkage and higher profits but also labor efficiency improvements that offset the costs of AI implementation [8]. Moreover, these initiatives prevented 26,705 tons of CO_2 emissions from reaching landfills, underscoring the significant environmental benefits. Another case study involving Winnow Solutions and IKEA showed that deploying AI to monitor and analyze food waste in IKEA's kitchens led to a 30 % reduction in food waste within just one year [59]. These examples demonstrate the potential of AI-driven solutions to predict food spoilage, optimize inventory management, and foster sustainable practices, significantly mitigating food waste across various settings.

8. AI/machine learning models for food spoilage prediction

Building on the case studies and successful applications of AI in reducing food waste, it is important to explore the underlying machine learning (ML) AI models that power food spoilage prediction and inventory management. The deployment of these AI models varies depending on the nature of the data, the complexity of the environment, and the specific objectives of the food system [60]. Understanding the strengths, limitations, and real-world applications of these models is critical to advancing AI's role in mitigating food waste.

Supervised learning models, such as Random Forest (RF), SVM, and

Artificial Neural Networks (ANN), play a foundational role in food spoilage prediction. By learning from historical, labelled data, these models help predict spoilage patterns based on variables like temperature, humidity, and microbiological conditions. For instance, RF is particularly adept at managing heterogeneous datasets, making it a solid choice for diverse food products [61]. However, its reliance on large, labelled datasets and difficulty with high-dimensional data highlight the challenges that can arise in real-world applications. SVM models, which excel at binary classification tasks [62], could be useful for distinguishing between fresh and spoiled food. They work by finding the optimal hyperplane that separates the two classes, making them suitable for such applications but can be computationally intensive for large-scale deployments. To address these challenges, researchers have explored various strategies to enhance the efficiency of SVMs. For instance, a study by Pouladzadeh et al. [63] discusses the use of SVMs in food recognition applications, which is a crucial step in calorie estimation. ANN models, especially deep learning networks, have shown great promise in detecting subtle spoilage indicators from complex datasets, such as hyperspectral imaging data, but come with higher computational costs and data requirements [64,65].

In contrast, unsupervised learning models, including K-Means Clustering and Principal Component Analysis (PCA), offer potential in anomaly detection and food spoilage prediction. These models do not require labelled data and can identify outliers in environmental conditions that may lead to spoilage. K-Means clustering, for example, helps segment food products based on spoilage characteristics, facilitating better inventory management [66]. However, challenges arise in handling non-spherical data distributions [67]. PCA, useful for dimensionality reduction, enhances computational efficiency but may overlook important spoilage indicators when critical features are discarded [68].

Deep learning models, including CNN and RNN, have revolutionized the prediction of food spoilage by analyzing image and time-series data. CNNs are particularly useful for visual quality assessments, such as detecting microbial contamination in fresh produce using hyperspectral imaging, enabling early detection of spoilage [69]. On the other hand, RNNs, particularly LSTM networks, excel in analyzing time-dependent data, such as temperature and humidity trends in cold storage facilities, predicting spoilage risks with high temporal accuracy [70,71]. Despite their advantages, CNNs and RNNs require large, labelled datasets and substantial computational resources, making them more challenging to implement for smaller-scale operations.

Reinforcement learning (RL) models, such as Deep Q-Networks (DQN), optimize inventory management by adjusting order patterns based on spoilage risks and demand fluctuations [72,73]. By dynamically learning the best decision-making strategies, RL can reduce food waste while maximizing profitability. Hybrid approaches, which integrate AI models with IoT sensors, provide real-time monitoring and enable adaptive control over food storage conditions. For instance, combining LSTM with IoT-based temperature sensors significantly enhances spoilage prediction accuracy by accounting for real-time environmental changes [74]. However, RL models require significant training time and continuous real-world feedback, presenting challenges for immediate scalability.

Despite the promising results demonstrated in case studies, the use of AI technologies to reduce food waste is not without its limitations and challenges. AI models heavily rely on large, high-quality datasets, which may not always be available, especially for smaller businesses with limited historical data. The initial costs of implementing AI systems, along with ongoing computational and maintenance requirements, can also be prohibitive. Additionally, while these technologies are scalable for large operations, their complexity and resource demands may make them difficult to apply in smaller-scale environments. Furthermore, AI solutions are often dependent on real-time data inputs, such as sensor readings, which can be vulnerable to inaccuracies or malfunctions. Issues related to model selection, computational efficiency, and potential job displacement also add to the complexity of widespread AI adoption in the food sector.

9. Future directions and research opportunities in AI for food spoilage prediction

Future research in AI and food spoilage prediction is poised to advance along several integrated and strategic paths to enhance the effectiveness and application of these technologies:

Technological advancements: Research efforts should focus on improving AI algorithms for better accuracy and reliability in predicting food spoilage. This involves developing advanced deep-learning models that can process and analyze complex datasets to pinpoint subtle spoilage indicators with greater precision.

Integration with emerging technologies: Investigating the synergies between AI and emerging technologies like the IoT and blockchain is crucial. IoT sensors offer continuous monitoring of environmental conditions and food quality parameters, feeding real-time data into AI algorithms for sharper spoilage predictions [75]. Similarly, blockchain technology enhances traceability and transparency within the food supply chain, aiding in the effective tracking of food products and enabling timely interventions to prevent spoilage [68]. This transparency enables AI systems to process data with greater precision tracking variables such as temperature fluctuations, humidity levels, and transit times which are critical to predicting spoilage [75]. By guaranteeing the integrity and availability of data throughout the entire food supply chain, blockchain facilitates the timely identification of spoilage risks and vulnerabilities. As a result, AI models can generate more reliable predictions and inform decision-making processes that lead to waste reduction, particularly by minimizing the risk of overstocking or mishandling perishable goods [33].

Multi-modal data fusion: The fusion of diverse data sources—including sensory, chemical, and microbiological data—into AI models can significantly enrich predictive accuracy and offer a deeper understanding of the mechanisms behind food spoilage. This comprehensive approach is essential for capturing the full spectrum of factors that influence spoilage.

Real-time monitoring and adaptive control: Developing AI systems equipped with real-time monitoring and adaptive control capabilities can enable proactive management of food storage conditions. By leveraging predictive insights, these systems can dynamically adjust variables such as storage environments, packaging materials, and distribution logistics to minimize waste and maximize food safety and quality [33].

Collaborative research initiatives: Fostering interdisciplinary collaborations among food scientists, data scientists, engineers, and industry stakeholders will accelerate the pace of innovation in AI applications for food spoilage prediction. Such collaborative efforts can harness diverse expertise and resources to tackle complex challenges and develop viable, sustainable solutions for managing food waste [76].

By focusing on these areas, researchers and industry practitioners can significantly advance the capabilities of AI in reducing food waste, with profound implications for both the economy and the environment.

10. Conclusion

AI has emerged as a transformative force in the battle against food waste and spoilage, unlocking new avenues for enhancing food safety and sustainability across the global supply chain. By harnessing advanced algorithms and real-time data analytics, AI enables the early detection of spoilage, optimizes inventory management, and significantly reduces waste through more accurate demand forecasting. The results are not just theoretical; companies like IKEA, Shelf Engine and Afresh have already demonstrated significant reduction in food waste per store, directly translating to meaningful environmental impacts, including the prevention of thousands of tons of CO_2 emissions. These

underscore AI's practical applicability and the potential for large-scale improvements in food systems.

The unique contribution of this study lies in its demonstration of AI's capacity to drive tangible, measurable change across both operational efficiencies and environmental outcomes. The results provide clear evidence that AI can revolutionize food waste management, offering solutions that not only reduce waste but also contribute to a more secure and sustainable global food supply. The ability to optimize resources, minimize waste, and reduce carbon footprints is essential for meeting the world's growing food demands while mitigating environmental harm.

To unlock AI's full potential, however, requires overcoming key challenges, such as data quality, model training, and integration with existing systems. Addressing these obstacles demands a collaborative effort among technologists, researchers, policymakers, and industry stakeholders. The integration of AI with complementary technologies like IoT and big data offers additional opportunities to create a smarter, more efficient food system. Future research must focus on overcoming the barriers of data privacy, model biases, and regulatory compliance, ensuring that AI's transformative potential is fully realized. By fostering interdisciplinary collaboration and standardizing data protocols, we can build a more sustainable, efficient, and resilient food supply chain that benefits both people and the planet for generations to come.

CRediT authorship contribution statement

Helen Onyeaka: Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization. **Adenike Akinsemolu:** Writing – review & editing, Writing – original draft. **Taghi Miri:** Writing – review & editing, Writing – original draft, Supervision. **Nnabueze Darlington Nnaji:** Writing – review & editing, Writing – original draft. **Keru Duan:** Writing – review & editing, Writing – original draft. **Gu Pang:** Writing – review & editing, Writing – original draft. **Phemelo Tamasiga:** Writing – review & editing, Writing – original draft. **Samran Khalid:** Writing – review & editing, Writing – original draft. **Zainab T. Al-Sharify:** Writing – review & editing, Writing – original draft. **Chinenye Ugwa:** Writing – review & editing, Writing – original draft.

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No data was used for the research described in the article.

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